SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA 0614 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks: 25**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Variganji Sumanth Babu / 192524447, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V. Sumanth Babu

Annexure A

Short Answer Test

1. With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme.

Answer: Machine-learning system blocks

A machine-learning (ML) system is an end-to-end process that turns data into a model and then uses that model to make predictions or discover groups in new data. The algorithm is only one block in the complete system; reliable results also depend on data quality, evaluation and monitoring.

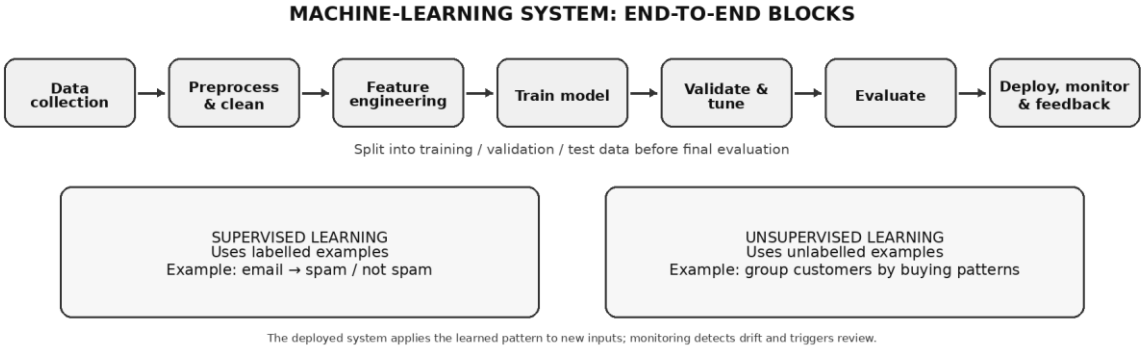


Figure 1. Major blocks of a machine-learning system and the two learning settings used in this answer.

Explanation of the blocks

1. Data collection and labelling. Collect relevant examples from sensors, databases, images, transactions or logs. In supervised learning, each example also needs a trusted target label.

2. Data storage and splitting. Store examples consistently and divide them into training, validation and test sets. Keep the test set separate until final evaluation to avoid an overly optimistic result.

3. Preprocessing. Remove duplicates, correct inconsistent values, handle missing data and outliers, encode categorical fields, and scale numerical values when the chosen algorithm needs it.

4. Feature engineering and selection. Convert raw data into useful input attributes and retain features that are relevant, measurable and available at prediction time. This can reduce noise and computation.

5. Model and algorithm selection. Choose a method suited to the task, such as a classifier for labelled categories, a regression model for numerical values, or clustering for unlabelled observations.

6. Training. The learning algorithm adjusts model parameters using the training examples to minimize a suitable loss or objective function.

7. Validation and tuning. Compare candidate models and adjust hyperparameters using validation data or cross-validation. The test set must not guide repeated tuning.

8. Evaluation. Use task-appropriate measures, such as precision, recall and F1-score for classification, or MAE/RMSE for regression. Inspect errors as well as a single score.

9. Deployment and monitoring. Expose the model to new inputs, monitor prediction quality and changes in incoming data, and retrain or review it when performance degrades.

Supervised versus unsupervised learning

Supervised learning uses labelled input-output pairs. For example, an email classifier learns from emails labelled “spam” or “not spam” and predicts a label for a new email. Unsupervised learning uses data without target labels and looks for structure; for example, clustering groups customers by similar purchase patterns. In short, supervised learning predicts a known target, while unsupervised learning discovers patterns that were not supplied as labels.

How the blocks work together

The blocks form a repeatable cycle rather than a one-time sequence. For a classification task, the model returns a category; for regression, it returns a numerical estimate; for clustering, it groups examples by similarity without a supplied target label. The feature format used in training must be reproduced exactly during prediction, otherwise the model receives inputs different from the ones it learned from. Evaluation should be selected to match the cost of mistakes: recall matters when missing a positive case is costly, while precision matters when false alarms are costly. Versioning the dataset, preprocessing steps and model makes it possible to reproduce results and investigate failures. These checks make the complete ML system more dependable than training an algorithm alone.

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using historical and real-time data. Explain the choice of model, data preprocessing, and feature selection techniques. (a) Describe the steps involved in training, validating, and evaluating the model. (b) Explain how the system can assist in early warning and emergency response.

Answer: Proposed disaster-management ML system

The system should combine historical disaster records with live sensor and geospatial data to estimate risk, assign a location-specific risk level and communicate uncertainty to emergency teams. It should support human decision-making, not automatically replace official warning authorities.

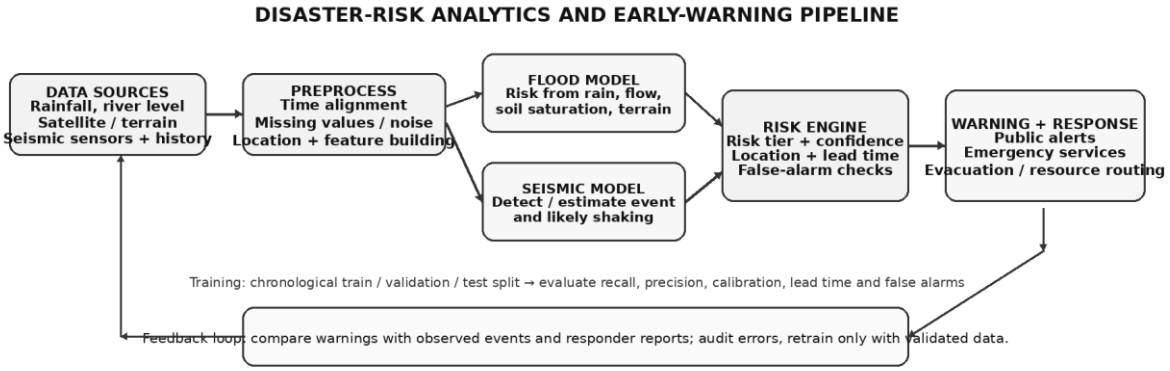


Figure 2. Data-to-warning architecture with separate flood and seismic analytics and a feedback loop.

1. Model choice and feature design

For structured flood-risk data, Random Forest or gradient-boosted trees are suitable starting models because they handle mixed numerical and categorical features, learn non-linear relationships and provide useful feature-importance estimates. When the main signal is a sequence over time, such as rainfall and river-level readings, a time-series model can be compared with the tree-based baseline. Seismic data needs a separate event-detection or shaking-estimation model using waveform and sensor features; it should not be presented as a reliable way to predict the exact time, place and magnitude of an earthquake in advance.

2. Data preprocessing and feature selection

1. Combine weather stations, river gauges, satellite/terrain data, historical flood reports, seismic sensors and verified event records using consistent timestamps and geographic coordinates.

2. Clean duplicates and impossible readings; handle missing values with a documented method; filter sensor noise; normalize units; encode categories; and align data to common time windows and map locations.

3. Construct features such as rainfall in the last 1, 6 and 24 hours, rate of river-level rise, soil saturation, elevation, slope, land cover and drainage proximity for floods. For seismic analysis, use measured waveform characteristics, signal amplitude, station coverage and event-detection features.

4. Select features using domain knowledge, correlation checks, permutation importance or cross-validated feature selection. Remove redundant variables and prevent leakage from information that would only be known after the event.

3. Training, validation and evaluation

Define the target carefully, for example flood occurrence or risk tier within a stated location and future time window. Split data chronologically so future records do not leak into training, and keep some regions or events for an independent generalization test. Train the model on the training set, select hyperparameters on validation data, then evaluate once on the untouched test set. Since disasters are rare but costly, report recall/sensitivity, precision, F1-score, calibration, false-alarm rate and warning lead time, not accuracy alone. Test across seasons and locations and review errors with domain experts.

4. Early warning and emergency response

When the estimated risk crosses a locally approved threshold, the system can send geotagged alerts, show risk maps, identify exposed communities and infrastructure, and help responders prioritize evacuation routes, shelters, rescue teams and supplies. Alerts should state the area, confidence, time window and recommended action; an authorized agency should verify and issue public warnings. Outcomes and responder feedback should be audited and used to improve the system.

To make the warning usable, the system should return more than a bare “yes/no” label. It should provide the affected geographic area, the forecast horizon, a risk tier, the confidence or uncertainty estimate, and the main signals behind the result. A dashboard can combine model output with flood maps and population or infrastructure layers, while a notification service distributes alerts to approved responders. Before deployment, run simulations and retrospective tests on historical events, including rare and severe cases, and document who is responsible for approving warnings. This operational layer is necessary because even a technically accurate model is not helpful if its output is late, unclear or disconnected from response procedures.

3. In the context of Find-S algorithm, consider the following dataset used to determine whether a day is suitable for playing tennis based on attributes such as Outlook, Temperature, Humidity, and Wind. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show the step-by-step updates of the hypothesis after each positive training example and ignore all negative examples. Finally, provide the final hypothesis.

Dataset given in the question

Example	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

Answer: Find-S algorithm

Find-S starts with the most specific hypothesis and generalizes it only when a positive example requires it. The hypothesis is written as h = ⟨Outlook, Temperature, Humidity, Wind⟩. The symbol Ø means “no attribute value accepted yet”; ? means “any value is accepted.” Negative examples are ignored exactly as instructed.

Training example	Update performed	Hypothesis after example
Initial	—	⟨Ø, Ø, Ø, Ø⟩
Example 1: No	Negative, ignore	⟨Ø, Ø, Ø, Ø⟩
Example 2: No	Negative, ignore	⟨Ø, Ø, Ø, Ø⟩
Example 3: Yes	First positive example	⟨Overcast, Hot, High, Weak⟩
Example 4: Yes	Outlook and temperature differ	⟨?, ?, High, Weak⟩
Example 5: Yes	Humidity differs (High/Normal)	⟨?, ?, ?, Weak⟩
Example 6: No	Negative, ignore	⟨?, ?, ?, Weak⟩
Example 7: Yes	Wind differs (Weak/Strong)	⟨?, ?, ?, ?⟩
Example 8: No	Negative, ignore	⟨?, ?, ?, ?⟩

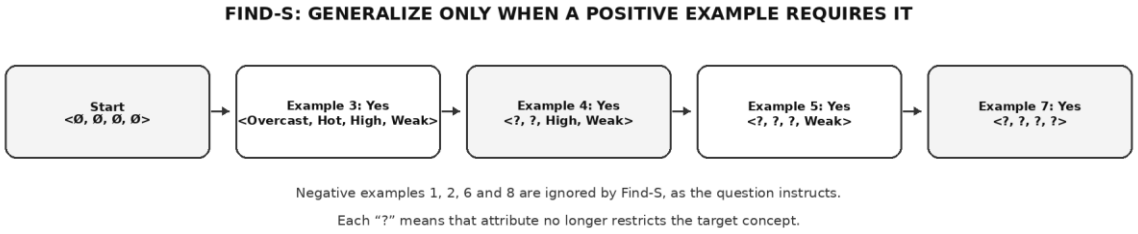


Figure 3. The hypothesis becomes more general as positive examples disagree on individual attributes.

Final hypothesis and interpretation

The final maximally specific hypothesis consistent with every positive example is h = ⟨?, ?, ?, ?⟩. Each attribute has to become unrestricted because the positive examples include more than one value for Outlook, Temperature, Humidity and Wind. Therefore Find-S would accept any combination of these four attributes. This outcome is very general and illustrates the limitation of ignoring negative examples: for instance, the dataset labels Example 8 as “No,” yet the all-? hypothesis would match it. Find-S produces a hypothesis from positive cases but does not guarantee consistency with the negative cases.

In this dataset, the positive rows successively remove constraints: Example 3 supplies the initial values, Example 4 forces Outlook and Temperature to become wildcards, Example 5 removes the Humidity constraint, and Example 7 removes the Wind constraint. A positive example can only make a Find-S hypothesis more general; it never adds a new restriction that would exclude a previously accepted positive instance. Since all four attributes vary across the positive cases, no specific value can remain in the final tuple. The answer follows from the given data and the instruction to ignore negative examples, rather than from a general claim about what weather makes tennis suitable.

4. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a person will enjoy a sport based on attributes such as Weather, Temperature, Humidity, and Wind. Using the Candidate Elimination algorithm, determine the version space by computing both the specific (S) and general (G) boundaries. Show the step-by-step updates after processing each training example.

Dataset given in the question

Example	Weather	Temperature	Humidity	Wind	Enjoy Sport
1	Sunny	Warm	Normal	Weak	Yes
2	Sunny	Warm	High	Strong	Yes
3	Rainy	Cold	High	Weak	No
4	Sunny	Cold	Normal	Weak	Yes
5	Rainy	Warm	Normal	Strong	No

Answer: Candidate Elimination

Candidate Elimination keeps two boundaries of the version space. S contains the most specific hypotheses still consistent with the observations; G contains the most general hypotheses still consistent with them. On a positive example, generalize S minimally and remove G hypotheses that reject that example. On a negative example, specialize inconsistent G hypotheses minimally so that they reject it while remaining at least as general as S.

Example	Observed values	S boundary after step	G boundary after step
Initial	—	⟨Ø,Ø,Ø,Ø⟩	{(?,?,?,?)}
1: Yes	Sunny, Warm, Normal, Weak	⟨Sunny,Warm,Normal,Weak⟩	{(?,?,?,?)}
2: Yes	Sunny, Warm, High, Strong	⟨Sunny,Warm,?,?⟩	{(?,?,?,?)}
3: No	Rainy, Cold, High, Weak	S unchanged	{⟨Sunny,?,?,?⟩, ⟨?,Warm,?,?⟩}
4: Yes	Sunny, Cold, Normal, Weak	⟨Sunny,?,?,?⟩	{⟨Sunny,?,?,?⟩}
5: No	Rainy, Warm, Normal, Strong	S and G unchanged	{⟨Sunny,?,?,?⟩}

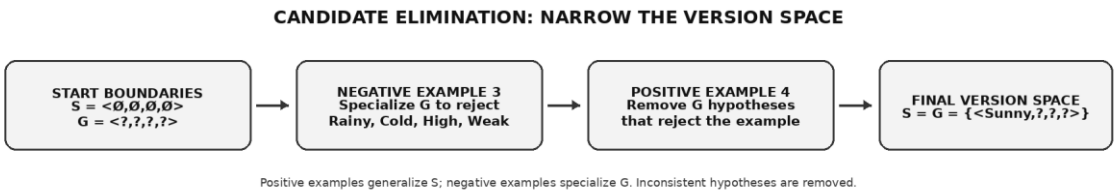


Figure 4. The negative third example temporarily splits G; the fourth positive example leaves only the Sunny boundary.

Result and check

After Example 3 is negative, the hypothesis ⟨?,?,?,?⟩ must be specialized. Sunny and Warm are the only values specified by the current S boundary that differ from the negative example, giving ⟨Sunny,?,?,?⟩ and ⟨?,Warm,?,?⟩. Example 4 is positive but does not satisfy ⟨?,Warm,?,?⟩ because Temperature is Cold, so that hypothesis is removed. The remaining S and G boundaries are identical; therefore the version space contains one hypothesis: ⟨Sunny,?,?,?⟩. The learned rule is that the person enjoys the sport on Sunny days regardless of Temperature, Humidity and Wind, within this dataset.

The boundary updates can be checked against each label. Examples 1 and 2 are positive, so S is generalized from the first complete instance to retain only Weather = Sunny and Temperature = Warm. Example 3 is negative; it conflicts with those two retained values, allowing two minimal specializations of G. Example 4 is positive and has Weather = Sunny but Temperature = Cold, so S must drop the Warm restriction, and G must drop the alternative that still requires Warm. The final negative example is Rainy and is already excluded by the remaining Sunny hypothesis. At every stage, a surviving specific hypothesis must be no more general than a surviving general hypothesis; otherwise the boundaries would not represent a valid version space. This is why the G candidate inconsistent with Example 4 is removed rather than kept as an alternative.

5. In the context of the Find-S algorithm, consider the dataset used to determine whether a customer will buy a product based on attributes such as Age, Income, Student, and Credit Rating. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show step-by-step updates after each positive example and ignore negative examples. Finally, provide the final hypothesis.

Dataset given in the question

Example	Age	Income	Student	Credit Rating	Buys Product
1	Young	High	No	Fair	No
2	Young	High	No	Excellent	No
3	Middle	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes

Answer: Find-S step-by-step

Represent a hypothesis as h = ⟨Age, Income, Student, Credit Rating⟩. Find-S begins at the most specific hypothesis ⟨Ø, Ø, Ø, Ø⟩. A negative example does not change the hypothesis, because the algorithm deliberately ignores negative instances.

Training example	Reason	Hypothesis after example
Initial	—	⟨Ø, Ø, Ø, Ø⟩
Example 1: No	Ignore negative instance	⟨Ø, Ø, Ø, Ø⟩
Example 2: No	Ignore negative instance	⟨Ø, Ø, Ø, Ø⟩
Example 3: Yes	First positive instance	⟨Middle, High, No, Fair⟩
Example 4: Yes	Age and Income differ; generalize them	⟨?, ?, No, Fair⟩

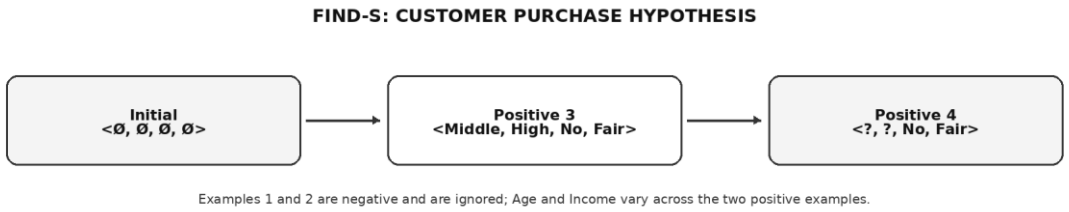


Figure 5. Only the two positive customer examples determine the final Find-S hypothesis.

Final hypothesis and interpretation

The maximally specific hypothesis consistent with the positive examples is h = ⟨?, ?, No, Fair⟩. Thus, according to the observed positive examples, the product is purchased when Student = No and Credit Rating = Fair, regardless of Age and Income. Age becomes ? because Middle and Senior differ; Income becomes ? because High and Medium differ. Student and Credit Rating remain fixed because both positive examples share the values No and Fair.

This is a hypothesis learned from only two positive cases, not proof that every customer with Student = No and Credit Rating = Fair will buy the product. The negative examples are not used to reject inconsistent generalizations in Find-S. A more complete assessment would compare the hypothesis against negative examples and validate it with more labelled customer data.

The table shows the exact point where the hypothesis changes. Before seeing a positive example, all four attributes are Ø because no positive instance has yet been accepted. Example 3 then initializes the hypothesis with the observed tuple. For Example 4, Find-S compares each attribute with the current tuple: Senior differs from Middle, so Age becomes ?; Medium differs from High, so Income becomes ?; Student remains No; Credit Rating remains Fair. The final tuple is therefore the most specific generalization that covers both positive examples under the conjunctive hypothesis representation. A wildcard is a deliberate allowance for all values, not an unknown or missing entry.

For a real purchasing system, this result would be too weak to use without further testing. The two negative examples are not used by Find-S, so the procedure cannot demonstrate that the final rule rejects them. A later evaluation should apply the learned rule to labelled records not used to construct it, count false positives and false negatives, and compare it with alternative models. If the dataset grows, cross-validation and a held-out test set can reveal whether the apparent pattern repeats for unseen customers. Find-S is useful here because its update steps are transparent, but its positive-only learning rule is also an important limitation.

6. In the context of the Find-S algorithm, consider a dataset used to determine whether a person will go for a workout based on attributes such as Weather, Temperature, Time, and Mood. Apply the Find-S algorithm and determine the maximally specific hypothesis. Show the hypothesis updates after each positive example only.

Dataset:

Example	Weather	Temperature	Time	Mood	Workout
1	Sunny	Hot	Morning	Tired	No
2	Rainy	Mild	Evening	Happy	Yes
3	Sunny	Mild	Evening	Happy	Yes
4	Cloudy	Cool	Morning	Happy	Yes

7. In the context of the Find-S algorithm, consider a dataset used to determine whether a loan should be approved based on attributes such as Employment Status, Salary Level, Credit Score, and Loan History. Apply the Find-S algorithm to derive the most specific hypothesis consistent with the positive examples.

Dataset:

Example	Employment	Salary	Credit Score	Loan History	Loan Approved
1	Employed	High	Good	Clean	Yes
2	Employed	Medium	Good	Clean	Yes
3	Unemployed	Low	Poor	Bad	No
4	Employed	Medium	Excellent	Clean	Yes

8. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a customer will subscribe to a service based on attributes such as Age Group, Income Level, Internet Usage, and Device Type. Determine the version space by computing the specific (S) and general (G) boundaries. Show step-by-step updates after each training example.

Dataset:

Example	Age Group	Income Level	Internet Usage	Device Type	Subscribe
1	Young	High	Frequent	Mobile	Yes
2	Young	High	Occasional	Mobile	Yes
3	Middle	Low	Frequent	Desktop	No
4	Young	Medium	Frequent	Mobile	Yes
5	Senior	High	Occasional	Desktop	No

9. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a patient follows a healthy lifestyle based on attributes such as Diet, Exercise, Sleep, and Stress Level. Apply the algorithm to compute the version space (S and G boundaries) and show the updates after each example.

Dataset:

Example	Diet	Exercise	Sleep	Stress Level	Healthy Lifestyle
1	Balanced	Regular	Adequate	Low	Yes
2	Balanced	Regular	Poor	Medium	Yes
3	Junk	None	Poor	High	No
4	Balanced	Moderate	Adequate	Low	Yes
5	Junk	Regular	Adequate	High	No

10. Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions.

11. Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples.

12. Design a machine learning model for spam email classification using labeled data.
(a) Explain the choice of algorithm (e.g., Naïve Bayes, SVM, or Logistic Regression).
(b) Describe the preprocessing steps such as tokenization, stop-word removal, and vectorization.
(c) Discuss evaluation metrics such as accuracy, precision, recall, and F1-score.

13. In the context of the Candidate-Elimination Algorithm, consider a dataset with attributes: Sky, AirTemp, Humidity, Wind, Water, Forecast and target concept EnjoySport.
(a) Explain the initialization of S (specific boundary) and G (general boundary).
(b) Show how S and G change with each training example.
(c) Compare Candidate-Elimination with the Find-S algorithm in terms of generalization capability.

14. Develop a machine learning system for predicting house prices using features like location, size, number of rooms, and amenities.
(a) Justify the choice of regression model (Linear Regression, Decision Tree, or Random Forest).
(b) Explain feature scaling and handling missing data.
(c) Describe how overfitting can be detected and avoided.

15. Explain the concept of clustering in unsupervised learning.
(a) Describe the working of the K-Means algorithm with a step-by-step example.
(b) Explain how the value of K is chosen using the Elbow Method.
(c) Compare K-Means with Hierarchical Clustering.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand